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ME 5130: Advanced Heat and Mass Transfer

Homework 01: Heat Conduction

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1–2, Problem Statement: Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as:

$$\nabla T = \hat{\mathbf{u}}_r \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} \quad (1)$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (q_\theta \sin \theta) \quad (2)$$

1–2, Solution: The transformation from spherical to cartesian coordinates is:

$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned} \quad (3)$$

and the following can be defined:

$$\begin{aligned} u_1 &= r \\ u_2 &= \phi \\ u_3 &= \theta \end{aligned} \quad (4)$$

Now, from Özışık, we have Eq. (1–19):

$$\nabla T = \hat{\mathbf{u}}_1 \frac{1}{a_1} \frac{\partial T}{\partial u_1} + \hat{\mathbf{u}}_2 \frac{1}{a_2} \frac{\partial T}{\partial u_2} + \hat{\mathbf{u}}_3 \frac{1}{a_3} \frac{\partial T}{\partial u_3} \quad (5)$$

Eq. (1–22a):

$$\nabla \cdot \mathbf{q} = \frac{1}{a} \left[\frac{\partial}{\partial u_1} \left(\frac{a}{a_1} q_1 \right) + \frac{\partial}{\partial u_2} \left(\frac{a}{a_2} q_2 \right) + \frac{\partial}{\partial u_3} \left(\frac{a}{a_3} q_3 \right) \right] \quad (6)$$

Eq. (1–22b):

$$a = a_1 a_2 a_3 \quad (7)$$

and Eq (1–18):

$$a_i^2 = \left(\frac{\partial x}{\partial u_i} \right)^2 + \left(\frac{\partial y}{\partial u_i} \right)^2 + \left(\frac{\partial z}{\partial u_i} \right)^2, \quad i = 1, 2, 3 \quad (8)$$

We can now compute the partial derivatives needed to find the scale factors:

$$\frac{\partial x}{\partial r} = \sin \theta \cos \phi \quad (9)$$

$$\frac{\partial y}{\partial r} = \sin \theta \sin \phi \quad (10)$$

$$\frac{\partial z}{\partial r} = \cos \theta \quad (11)$$

$$\frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi \quad (12)$$

$$\frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi \quad (13)$$

$$\frac{\partial z}{\partial \phi} = 0 \quad (14)$$

$$\frac{\partial x}{\partial \theta} = r \cos \theta \cos \phi \quad (15)$$

$$\frac{\partial y}{\partial \theta} = r \cos \theta \sin \phi \quad (16)$$

$$\frac{\partial z}{\partial \theta} = -r \sin \theta \quad (17)$$

We can now evaluate:

$$\begin{aligned} a_r^2 &= \sin^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \theta \\ &= \sin^2 \theta (\cos^2 \phi + \sin^2 \phi) + \cos^2 \theta \\ &= \sin^2 \theta + \cos^2 \theta \\ &= 1 \end{aligned} \quad (18)$$

$$a_1 = a_r = 1$$

$$\begin{aligned} a_\phi^2 &= r^2 \sin^2 \theta \sin^2 \phi + r^2 \sin^2 \theta \cos^2 \phi \\ &= r^2 \sin^2 \theta (\sin^2 \phi + \cos^2 \phi) \\ &= r^2 \sin^2 \theta \end{aligned} \quad (19)$$

$$a_2 = a_\phi = r \sin \theta$$

$$\begin{aligned} a_\theta^2 &= r^2 \cos^2 \theta \cos^2 \phi + r^2 \cos^2 \theta \sin^2 \phi + r^2 \sin^2 \theta \\ &= r^2 \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + r^2 \sin^2 \theta \\ &= r^2 \cos^2 \theta + r^2 \sin^2 \theta \\ &= r^2 (\cos^2 \theta + \sin^2 \theta) \\ &= r^2 \end{aligned} \quad (20)$$

$$a_3 = a_\theta = r$$

We should note that since spherical coordinates (r, ϕ, θ) form an orthogonal coordinate system, we can use equations (1–19) and (1–22). Nevertheless, we can now re-write equation (1–19):

$$\nabla T = \hat{\mathbf{u}}_r \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_\phi \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_\theta \frac{1}{r} \frac{\partial T}{\partial \theta} \quad (21)$$

We can also now write:

$$a = a_1 a_2 a_3 = a_r a_{phi} a_\theta = r^2 \sin \theta \quad (22)$$

Therefore:

$$\frac{a}{a_r} = r^2 \sin \theta \quad (23)$$

$$\frac{a}{a_\phi} = r \quad (24)$$

$$\frac{a}{a_\theta} = r \sin \theta \quad (25)$$

Therefore, equation (1-22a) can be re-written as:

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2 \sin \theta} \left[\frac{\partial}{\partial r} (r^2 \sin \theta q_r) + \frac{\partial}{\partial \phi} (r q_\phi) + \frac{\partial}{\partial \theta} (r \sin \theta q_\theta) \right] \quad (26)$$

Which can be simplified to:

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 q_r) + \frac{1}{r \sin \theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta q_\theta) \quad (27)$$

1-12, Problem Statement: Set up the mathematical formulation of the following head conduction problems:

1-12, Solution:

1. A slab in $0 \leq x \leq L$ is initially at a temperature $F(x)$. For times $t > 0$, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ dissipates heat by convection into a medium at zero temperature.

For a slab where $T(x, 0) = F(x)$ with no generation, the heat equation can be written as:

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}, \quad 0 < x < L, t > 0 \quad (28)$$

At the $x = 0$ boundary condition:

$$\frac{\partial T}{\partial x} = 0, \quad x = 0, t > 0 \quad (29)$$

At the $x = L$ boundary condition:

$$\begin{aligned}
-k \frac{\partial T}{\partial x} &= h(T - T_{infy}) \\
-k \frac{\partial T}{\partial x} &= hT \\
k \frac{\partial T}{\partial x} + hT &= 0, \quad x = L, t > 0
\end{aligned} \tag{30}$$

and finally, at the initial condition $t = 0$:

$$T(x, 0) = F(x), \quad 0 \leq x \leq L, t = 0 \tag{31}$$

2. A semiinfinite region $0 \leq x < \infty$ is initially at a temperature $F(x)$. For times $t > 0$, heat is generated in the medium at a constant rate of $g_0 W/m^3$, while the boundary at $x = 0$ is kept at zero temperature.

For a semiinfinite region with a constant rate of heat generation of $g_0 W/m^3$, the heat equation can be written as:

$$\frac{\partial^2 T}{\partial x^2} + \frac{g_0}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}, \quad 0 < x < \infty, t > 0 \tag{32}$$

At the $x = 0$ boundary condition:

$$T(x, t) = 0, \quad x = 0, t > 0 \tag{33}$$

and finally, at the initial condition $t = 0$:

$$T(x, 0) = F(x), \quad 0 \leq x \leq \infty, t = 0 \tag{34}$$

3. A solid cylinder $0 \leq r \leq b$ is initially at a temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ dissipates heat by convection into a medium at zero temperature.

For a solid cylinder with heat generated at a rate of $g(r)$, the heat equation can be written as:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{g(r)}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}, \quad 0 \leq r < b, t > 0 \tag{35}$$

At the boundary condition $r = 0$, by symmetry:

$$\frac{\partial T}{\partial r} = 0, \quad r = 0, t > 0 \tag{36}$$

At the boundary condition $r = b$:

$$\begin{aligned}
-k \frac{\partial T}{\partial r} &= h(T - T_\infty) \\
-k \frac{\partial T}{\partial r} &= hT \\
k \frac{\partial T}{\partial r} + hT &= 0, \quad r = b, t > 0
\end{aligned} \tag{37}$$

and finally, at the initial condition $t = 0$:

$$T(r, 0) = F(r), \quad 0 \leq r \leq b, t = 0 \tag{38}$$

4. A solid sphere $0 \leq r \leq b$ is initially at temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3 , while the boundary at $r = b$ is kept at a uniform temperature T_0 .

For a solid sphere with heat generated at a rate of $g(r)$, the heat equation can be written as:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{g(r)}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}, \quad 0 \leq r < b, t > 0 \tag{39}$$

At the boundary condition $r = 0$, by symmetry:

$$\frac{\partial T}{\partial r} = 0, \quad r = 0, t > 0 \tag{40}$$

At the boundary condition $r = b$:

$$T(b, t) = T_0, \quad r = b, t > 0 \tag{41}$$

and finally, at the initial condition $t = 0$:

$$T(r, 0) = F(r), \quad 0 \leq r \leq b, t = 0 \tag{42}$$

1–15, Problem Statement: A long cylindrical iron bar of diameter $D = 5$ cm, initially at temperature $T_0 = 650^\circ\text{C}$, is exposed to an air stream at $T_\infty = 50^\circ\text{C}$. The heat transfer coefficient between the air stream and the surface of the bar is $h = 80 \text{ W m}^{-2} \text{ }^\circ\text{C}^{-1}$. Thermophysical properties may be taken as $\rho = 7800 \text{ kg m}^{-3}$, $C_p = 460 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$, and $k = 60 \text{ W m}^{-1} \text{ }^\circ\text{C}^{-1}$. Determine the time required for the temperature of the bar to reach 250°C by using the lumped system analysis.

1–15, Solution:

1–16, Problem Statement: A thermocouple is to be used to measure the temperature in a gas stream. The junction may be approximated as a sphere having thermal conductivity $k = 25 \text{ W m}^{-1} \text{ }^\circ\text{C}^{-1}$, $\rho = 8400 \text{ kg m}^{-3}$, and $C_p = 0.4 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1}$. The heat transfer coefficient

between the junction and the gas stream is $h = 560 \text{ W m}^{-2} \text{ }^{\circ}\text{C}^{-1}$. Calculate the diameter of the junction if the thermocouple should measure 95% of the applied temperature difference in 3s.

1–16, Solution:

2–2, Problem Statement: A slab, $0 \leq x \leq L$, is initially at a temperature $F(x)$. For times $t > 0$ the boundary surface at $x = 0$ is kept insulated and that at $x = L$ dissipates heat by convection into a medium at zero temperature with a heat transfer coefficient h . Obtain an expression for the temperature distribution $T(x, t)$ in the slab for times $t > 0$ and for the heat flux at the boundary surface $x = L$.

2–2, Solution:

2–6, Problem Statement: In a one-dimensional infinite medium, $-\infty < x < \infty$, initially, the region $a < x < b$ is at a constant temperature T_0 , and everywhere outside this region is at zero temperature. Obtain an expression for the temperature distribution $T(x, t)$ in the medium for times $t > 0$.

2–6, Solution:

2–15, Problem Statement: Obtain an expression for the steady-state temperature distribution $T(x, y)$ in a rectangular region $0 \leq x \leq a$, $0 \leq y \leq b$ for the following boundary conditions: the boundary at $x = 0$ is kept insulated, the boundary at $y = 0$ is kept at temperature $f(x)$ and the boundaries at $x = a$ and $y = b$ dissipate heat by convection into an environment at zero temperature. Assume the heat transfer coefficient to be the same for both boundaries.

2–15, Solution: